

Physik II — FS 2024 — Prof. R. Wallny — Serie 8

Solution

Exercise 1: Muons

- a) In the following, the time in the reference frame of the muons will be labeled t' and the time in our reference frame (that of the Earth) will be labeled t . In the reference frame of the muons we have:

$$n(t') = n_0 e^{-\alpha t'} \quad (1)$$

Considering that $\alpha = \frac{1}{\tau}$ the equation (1) becomes:

$$n(t') = n_0 e^{-\frac{t'}{\tau}} \quad (2)$$

where $\tau = 2.2 \mu s$ (in the reference frame of the muons).

In our reference frame: $n(t) = n_0 e^{-\frac{t}{\gamma\tau}}$ where $\gamma = \left(1 - \frac{v^2}{c^2}\right)^{-\frac{1}{2}} \simeq 7.09$ with $v = 0.99c$.

Considering h in our inertial frame: $t = \frac{h}{0.99c}$

$$\rightarrow \underbrace{\frac{n(t)}{n_0}}_{=1\%} = e^{-\left(\frac{h}{v}\right) \frac{1}{\gamma\tau}} \quad (3)$$

$$\ln(0.01) = -\left(\frac{h}{v}\right) \frac{1}{\gamma\tau} \quad (4)$$

$$h = -\ln(0.01) \cdot v \cdot \gamma\tau \quad (5)$$

$$\simeq 21.3 \text{ km} \quad (6)$$

The altitude is: $h \simeq 21.3 \text{ km}$.

- b) In the reference frame of the muon $h' = \gamma^{-1} h$. After insertion this gives $h' \simeq 3.0 \text{ km}$
- c) Without relativistic effects $h' = h$, or $\gamma = 1$ hold. Therefore, the number of muons that have not decayed yet is given by $\frac{n(t)}{n_0} = e^{-\frac{t'}{\tau}} \simeq 10^{-12} \%$. Without time dilation, as good as no muons would reach the earth.

Exercise 2: Non-commutativity of Lorentz-Poincaré transformations

- a) We know from the lecture that in the relativistic range it is easier to represent vectors as the space-time vectors of the form:

$$r = \begin{pmatrix} ct \\ x \\ y \\ z \end{pmatrix}$$

Thus we can define

$$\beta = v/c$$

and

$$\gamma = 1/\sqrt{(1 - \beta^2)}$$

If we consider the transformation from R to R' with a constant speed v along the x -axis the components of this vector can be written as:

$$r' = \begin{pmatrix} ct' \\ x' \\ y' \\ z' \end{pmatrix} = \begin{pmatrix} \gamma(ct - \beta x) \\ \gamma(x - \beta ct) \\ y \\ z \end{pmatrix}$$

So the transformation from R to R' comes as:

$$\begin{pmatrix} ct' \\ x' \\ y' \\ z' \end{pmatrix} = \Lambda(\vec{v}_1) \begin{pmatrix} ct \\ x \\ y \\ z \end{pmatrix}$$

with

$$\Lambda(\vec{v}_1) = \begin{bmatrix} \gamma & -\beta\gamma & 0 & 0 \\ -\beta\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (7)$$

b) The same way, we obtain the transformation from R' to R'' for a relative speed along the y -axis:

$$\begin{pmatrix} ct'' \\ x'' \\ y'' \\ z'' \end{pmatrix} = \Lambda(\vec{v}_2) \begin{pmatrix} ct' \\ x' \\ y' \\ z' \end{pmatrix}$$

with

$$\Lambda(\vec{v}_2) = \begin{bmatrix} \gamma & 0 & -\beta\gamma & 0 \\ 0 & 1 & 0 & 0 \\ -\beta\gamma & 0 & \gamma & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (8)$$

c) At first glance there are two composed transformations since we can form two changes of basis matrices, which are $\Lambda(v_2)\Lambda(v_1)$ and $\Lambda(v_1)\Lambda(v_2)$. These differ in the order in which the transformations are performed.

$$\Lambda(\vec{v}_1)\Lambda(\vec{v}_2) = \begin{bmatrix} \gamma^2 & -\beta\gamma & -\beta\gamma^2 & 0 \\ -\beta\gamma^2 & \gamma & \beta^2\gamma^2 & 0 \\ -\beta\gamma & 0 & \gamma & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (9)$$

and

$$\Lambda(\vec{v}_2)\Lambda(\vec{v}_1) = \begin{bmatrix} \gamma^2 & -\beta\gamma^2 & -\beta\gamma & 0 \\ -\beta\gamma & \gamma & 0 & 0 \\ -\beta\gamma^2 & \beta^2\gamma^2 & \gamma & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \quad (10)$$

It appears that $\Lambda(v_2)\Lambda(v_1) \neq \Lambda(v_1)\Lambda(v_2)$. Thus, the order of the transformations matters. The Lorentz- transformations are not commutative.

- d) Translations in three-dimensional, non-relativistic space are commutative. According to equation (5.20) in the script they are described by the following matrix:

$$\mathbb{M}_G^{\mu\nu} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -\beta^1 & 1 & 0 & 0 \\ -\beta^2 & 0 & 1 & 0 \\ -\beta^3 & 0 & 0 & 1 \end{bmatrix}. \quad (11)$$

In contrast to (7) and (8), this matrix shows two decisive differences. Firstly, the time is not influenced by any component. Secondly, all space components are only influenced by the time component. The space components can not influence each other. Hence, none of these transformations can influence another one. However, in three dimensional euclidian space there exist transformations which do not fullfil this condition, e.g. rotation. The matrix for rotation around the z -axis is given by

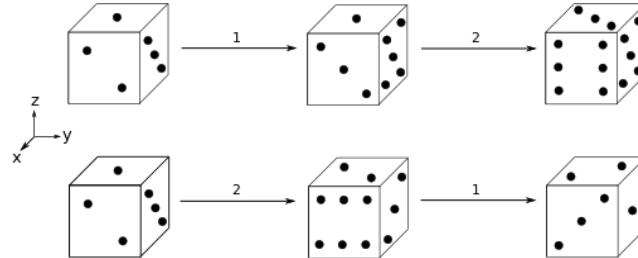
$$\mathbb{D}_z(\alpha) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \alpha & \sin \alpha & 0 \\ 0 & -\sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \quad (12)$$

The components x^1 and x^2 of this matrix influence each other. This becomes evident when calculating $x' = \cos \alpha x + \sin \alpha y$. Therefore, barely another transformation which is based on these components or changes these, is going to commute.

We consider a dice and two transformations as an example:

- A 90° rotation clockwise around the z -axis
- A 90° rotation clockwise around the y -axis

The figure shows both rotations in different order of performance. Evidently, the results differ.



Note: Transformations with relativistic velocities have the matrix form of rotations in the Minkowski space. The determinante of $\Lambda(\beta)$ is already 1. When introducing the rapidity $\theta = \arctan \beta$, $\Lambda(\theta)$ is additionally orthogonal and thus an ordinary rotation matrix.

e) If we assume that $v \ll c$ then β goes to zero. The Taylor development of γ around zero is

$$\gamma(\beta)_0 = 1 + \beta^2/2 + O(\beta^3) . \quad (13)$$

There is no linear term and we neglect terms from second order. Therefore, we use $\gamma(\beta)_0 \approx 1$.

We calculate again the transformations from question a) and b). We obtain:

$$\Lambda(\vec{v}_1^0) = \begin{bmatrix} 1 & -\beta & 0 & 0 \\ -\beta & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (14)$$

and,

$$\Lambda(\vec{v}_2^0) = \begin{bmatrix} 1 & 0 & -\beta & 0 \\ 0 & 1 & 0 & 0 \\ -\beta & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} . \quad (15)$$

If we now calculate the combined transformations $\Lambda(\vec{v}_1^0)\Lambda(\vec{v}_2^0)$ or $\Lambda(\vec{v}_2^0)\Lambda(\vec{v}_1^0)$ we obtain the following matrices:

$$\Lambda(\vec{v}_1^0)\Lambda(\vec{v}_2^0) = \begin{bmatrix} 1 & -\beta & -\beta & 0 \\ -\beta & 1 & \beta^2 & 0 \\ -\beta & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (16)$$

$$\Lambda(\vec{v}_2^0)\Lambda(\vec{v}_1^0) = \begin{bmatrix} 1 & -\beta & -\beta & 0 \\ -\beta & 1 & 0 & 0 \\ -\beta & \beta^2 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (17)$$

These still differ, but only by two entries of β^2 . β^2 is much smaller than β and therefore negligible in equation (13). We can also do that here and obtain the same matrix.

$$\Lambda(\vec{v}_1^0, \vec{v}_2^0) \approx \begin{bmatrix} 1 & -\beta & -\beta & 0 \\ -\beta & 1 & 0 & 0 \\ -\beta & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (18)$$

In the non-relativistic case these transformations are commutative as expected.

Exercise 3: Time dilation

- a) We denote the emission of the first and of the second signal as events A and B , respectively. In the reference frame S' of the spaceship, they have the space-time-coordinates (x'_A, t'_A) and $(x'_B = x'_A, t'_B = t'_A + \Delta T')$, respectively. In the reference frame S of Earth, events A and B have the coordinates (x_A, t_A) and $(x_B = x_A + \Delta x, t_B = t_A + \Delta t)$, respectively. The transformation from S' to S is given by:

$$x = \gamma(x' + vt'), \quad t = \gamma(t' + \frac{v}{c^2}x'), \quad (19)$$

with $\gamma = 1/\sqrt{1 - v^2/c^2} = 2$.

Therefore we have:

$$x_A = \gamma(x'_A + vt'_A), \quad t_A = \gamma(t'_A + \frac{v}{c^2}x'_A), \quad (20)$$

$$x_B = \gamma(x'_B + vt'_B), \quad t_B = \gamma(t'_B + \frac{v}{c^2}x'_B), \quad (21)$$

and further, with $x'_B - x'_A = 0$:

$$x_B - x_A = \Delta x = \gamma v \Delta t' \equiv \gamma v \Delta T', \quad t_B - t_A = \Delta t = \gamma \Delta t' \equiv \gamma \Delta T'. \quad (22)$$

In S , the time separation of the two events is $\Delta t = \gamma \Delta T'$. During this time, the spaceship travels the distance Δx . The two light signals arrive at the point x_0 on Earth at the times T_A and $T_B = T_A + \Delta T$, respectively. T_A and T_B can be calculated from:

$$T_A = t_A + \frac{x_A - x_0}{c}, \quad T_B = t_B + \frac{x_B - x_0}{c}, \quad (23)$$

where $(x_A - x_0)/c$ and $(x_B - x_0)/c$ are the delays (in S) of the signals from point x_A respectively x_B to x_0 .

Thus:

$$T_B - T_A = \Delta T = t_B - t_A + \frac{1}{c}(x_B - x_A) = \Delta t + \frac{1}{c}\Delta x, \quad (24)$$

i.e., the measured time difference between the signals consists of the time difference Δt in S (between the emission of the signals) and the difference in signal delay. With the above equations we get:

$$\Delta T = \gamma(1 + \frac{v}{c})\Delta T' = \frac{(1 + v/c)\Delta T'}{\sqrt{(1 + v/c)(1 - v/c)}} = \sqrt{\frac{1 + v/c}{1 - v/c}}\Delta T'. \quad (25)$$

With $\Delta T' = 4$ s we get $\Delta T = 15$ s. Compare this result with the relativistic Doppler effect.

- b) The distance Δx covered by the spacecraft (as seen from Earth) between the emissions of the signals is $\Delta x = \gamma v \Delta T' = 2.1 \cdot 10^9$ m.
- c) The body has the kinetic energy $E_{\text{kin}} = E_{\text{tot}} - E_{\text{rest}} = (m\gamma - m)c^2 = 9 \cdot 10^{16}$ J.

Exercise 4: Twin paradox using the Doppler effect

- a) When the space ship is moving away from Earth the frequencies of the signals received by both A and B' are shifted to lower frequencies as

$$f_L = f_0 \sqrt{\frac{1 - \beta}{1 + \beta}} \quad (\text{low rate signals}) \quad (26)$$

according to the Doppler-effect where $\beta \equiv \frac{v}{c}$ and $f_0 = 1$ s/y (signal/year) is the proper (measured in the reference frame of emitter) frequency of the radio emitters. Similarly, when the space ship is approaching Earth, the frequencies of the signals received for both observers are Doppler shifted to higher values as

$$f_H = f_0 \sqrt{\frac{1 + \beta}{1 - \beta}} \quad (\text{high rate signals}) \quad (27)$$

The low rate signals are from the receding ship and the high rate signals are from the approaching ship.

As seen by brother A on Earth

The number of signals from B' received by A on Earth is the sum of the low rate and high rate successions of signals.

The time T_L on Earth for the incoming low rate signals consists of two intervals: the time

$$T_1 = L_0/v \equiv t_0, \quad (28)$$

during which these signals are produced by the receding source and the time

$$T_2 = L_0/c = \beta t_0 \quad (29)$$

needed for last of these signals to travel the distance L_0 between the Earth and the source. So the total time for low rate signals is

$$T_L = T_1 + T_2 = (1 + \beta)t_0. \quad (30)$$

Note that L_0 is the distance Earth- α -Centauri as measured from the Earth reference system.

The amount of low rate signals measured by A is thus

$$N_L = f_L \cdot T_L = f_0 \sqrt{\frac{1 - \beta}{1 + \beta}} \cdot T_L = f_0 \cdot t_0(1 + \beta) \sqrt{\frac{1 - \beta}{1 + \beta}} = \frac{f_0 t_0}{\gamma(v)}. \quad (31)$$

The time T_H of high rate signals coming to Earth from the ship is just the time interval between the moment of the first such signal's arrival and the moment of the ship's return to the Earth:

$$T_H = \frac{L_0}{v} - \frac{L_0}{c} = t_0(1 - \beta). \quad (32)$$

The corresponding number of high rate signals is

$$N_H = f_0 \sqrt{\frac{1 + \beta}{1 - \beta}} \cdot T_H = \frac{t_0 f_0}{\gamma(v)}. \quad (33)$$

The total number of signals N'_{tot} emitted from the ship (which are equal to the number of signal received on earth) is

$$N'_{\text{tot}} = N_L + N_H = 2 \frac{f_0 t_0}{\gamma(v)}. \quad (34)$$

The number of years that the twin B' passed on the trip according to A is thus $t' = 2 \frac{t_0}{\gamma(v)}$. We used the fact that $f_0 = 1$ signal/year.

For brother A the time elapsed is simply given by $t = 2L_0/v = 2t_0$. Therefore, for brother A , the astronaut B' will be younger as $\gamma(v) > 1$.

As seen by brother B' on the space ship

In a similar way we calculate the number of signals emitted from Earth and received on the ship. During the first part of the journey, the ship's detector counts low rate signals f_L according to equation (26). During the second part, which starts immediately after the turn, the ship receives high rate signals f_H according to equation (27). The time needed for the space ship to travel from Earth to α -Centauri in the ship reference frame is given by the distance the ship has to travel $L' = \frac{L_0}{\gamma(v)}$ (length contraction) divided by the velocity with which the star is approaching. The time needed to reach α -Centauri is thus $L'/v = t_0/\gamma(v)$. The same time is needed for the space ship to travel from α -Centauri to the Earth.

Therefore, the total number of signals (low and high rate counts) in the ship's detectors is

$$N_{\text{tot}} = \frac{t_0 f_0}{\gamma(v)} \left(\sqrt{\frac{1-\beta}{1+\beta}} + \sqrt{\frac{1+\beta}{1-\beta}} \right) = 2f_0 t_0. \quad (35)$$

b) The number of years that the twin A spent on Earth according to B' is thus $2t_0$.

For brother B' the time elapsed is simply given by $2t_0/\gamma(v)$. From this we can conclude that, for brother B' , the astronaut A will be older, which is consistent with what A measures.

Exercise 5: Velocity measurement using radar signals

We calculate in the reference system of Earth. We assume that the two reference systems (of Earth and of the rocket) at $t = 0$ have common origins. The first beam is reflected by the first mirror (back mirror) at $t = 0$. The time Δt_2 needed by the second beam to reach the second mirror (front mirror) is given by

$$\Delta t_2 = \frac{d/\gamma + v\Delta t_2}{c}. \quad (36)$$

The first term accounts for the Lorentz contraction of the mirror distance d for an observer on Earth. The second term accounts for the motion of the rocket. The speed of light c does not depend on the reference system. Solving this equation for Δt_2 we get

$$\Delta t_2 = \frac{d}{\gamma(c-v)}. \quad (37)$$

After the reflection, the beam travels back to Earth, needing the same time Δt_2 . Hence

$$\Delta t = 2\Delta t_2 = \frac{2d\sqrt{1-v^2/c^2}}{c-v}, \quad (38)$$

where $\gamma = 1/\sqrt{1 - v^2/c^2}$ was used. Now we have to solve this equation for v . At first we multiply it with c and square it:

$$(c\Delta t)^2 = (2d)^2 \frac{c^2 - v^2}{(c - v)^2}, \quad (39)$$

$$= (2d)^2 \frac{c + v}{c - v}, \quad (40)$$

where $c^2 - v^2 = (c + v)(c - v)$ was used. It follows

$$(c\Delta t)^2(c - v) = (2d)^2(c + v). \quad (41)$$

Solving for v gives the final result:

$$\frac{v}{c} = \frac{(c\Delta t)^2 - (2d)^2}{(c\Delta t)^2 + (2d)^2}. \quad (42)$$

ALTERNATIVE SOLUTION:

We choose the two reference systems (system Earth and system rocket) in a way that they have a common origin at $t = 0$. The first beam is reflected by the first (backfacing) mirror at $t = 0$. In the coordinate frame of the rocket the second (front) mirror is located at the position $X'_1 = d$. The laser pulse reaches this mirror at time $T'_1 = \frac{d}{c}$. These space-time coordinates can now be transformed into the system Earth:

$$cT_1 = c\gamma \left(T'_1 + \frac{v}{c^2} X'_1 \right) = \gamma d(1 + \beta), \quad (43)$$

$$X_1 = \gamma(X'_1 + vT'_1) = \gamma d(1 + \beta). \quad (44)$$

Here, the coordinate systems have been chosen in a way that the terms in the brackets are positive in order to avoid a negative β . In the coordinate system of Earth the laser pulse takes an additional time from $T_2 = \frac{X_1}{c}$ to return to the origin of the coordinate system. Therewith, the time between the reflected pulses is to be determined using

$$\Delta t = T_1 + T_2 = \frac{2d}{c} \gamma(1 + \beta) = \frac{2d}{c} \frac{1 + \beta}{\sqrt{1 - \beta^2}} = \frac{2d}{c} \sqrt{\frac{1 + \beta}{1 - \beta}} \quad (45)$$

where we inserted $\gamma = \frac{1}{\sqrt{1 - \beta^2}}$, and used $1 - \beta^2 = (1 - \beta)(1 + \beta)$. We now want to solve this equation for β by first multiplying by c and then taking the square:

$$c^2 \Delta t^2 = 4d^2 \frac{1 + \beta}{1 - \beta} \quad (46)$$

This can be reshaped into

$$c^2 \Delta t^2 (1 - \beta) = 4d^2 (1 + \beta). \quad (47)$$

Solving for v gives

$$v = c \cdot \beta = c \cdot \frac{(c\Delta t)^2 - (2d)^2}{(c\Delta t)^2 + (2d)^2}. \quad (48)$$